A Paired Causal Audit of Negatively Curated Memory in a Llama-3 Trading Replay

Corrected pre-submission manuscript - fully logged paired controls and separate mechanism audit

Abstract

Memory-augmented language agents are often assumed to improve sequential decisions, but causal tests of particular memory configurations remain rare. We present a paired audit protocol for one local `llama3:latest` model in a teacher-forced SPY replay. The study freezes 32 RSI-crossing setups, delays retrieval until a prior trade's ten-day outcome has matured, evaluates every setup in every arm, and saves the full paired prompt and response trace. The four direct conditions separately manipulate an explicit trauma frame, original reflection prose, negative numerical records, and a padded no-memory control. A second 2 x 2 auxiliary prompt experiment independently manipulates a numeric confidence score (30/100 or 70/100) and a categorical status tag (NEUTRAL or SHAKEN). The loss-trauma arm produced 0 ABORTs in 32 setups (0.0%). All four direct arms produced the same action on every setup; the auxiliary factorial cells likewise produced 96 EXECUTE decisions. The exploratory draft reported 11 ABORTs in 34 setups; this corrected run therefore does not reproduce that earlier behavior, although the data snapshot and prompt protocol are not identical. An explicit hard-risk positive control produced ABORT on 24/24 eligible cases, confirming that the harness can register a deviation. The configuration-level random-partition benchmark was undefined because all decisions took the same action. These results are a configuration-specific prompt audit, not a live-trading result, an equivalence claim, or evidence about LLMs generally.

1. Introduction

Episodic memory is a common component of language-agent architectures. In financial-agent settings, however, a memory can contain both numerical outcomes and semantically charged language, and its retrieval policy can be highly selective. A binary comparison of a loss-memory prompt against an invented positive filler therefore cannot identify whether a decision change follows from the outer frame, the reflection language, the numerical losses, or the filler itself.

This paper reports a deliberately narrow experiment: a paired causal audit of one negatively curated retrieval configuration in one local model. Rather than assume that an observed abort is caused by memory, we evaluate every setup under every controlled prompt condition and report both directions of decision switches. Rather than infer a language-versus-number mechanism from correlated labels, we manipulate the categorical label and numerical score independently.

2. Related Work

Memory and reflection are widely used to make language agents persistent and adaptive (Park et al., 2023; Shinn et al., 2023; Sumers et al., 2024). In financial-agent work, FinMem uses layered memory and trader-character design to support decisions (Yu et al., 2023), while FinAgent combines diversified retrieval with multi-level reflection in a larger trading architecture (Zhang et al., 2024). Li et al. (2026) study behavioral consistency and switching in a financial stock-market simulation. These systems motivate a causal audit of a memory configuration, but they do not make a negative-only, trauma-framed retrieval policy representative of memory architectures in general. Our contribution is therefore a narrow protocol and trace audit, not a comparison with or refutation of those systems.

3. Protocol

Data and replay timing

We downloaded daily SPY OHLCV data from 2018-01-01 through 2023-12-31 and froze it before the model run (SHA-256: b58919ba5cf55a4529e96a0a0b90a2426cd4d6e79682636e02d1d3bc8ee8e768). The baseline signal is a 14-day RSI crossing from at least 40 to below 40 after 2021-01-01. A setup's outcome is its ten-trading-day forward return. Unlike the exploratory version, a reflection is eligible for retrieval only after its outcome date; a later decision never receives a reflection whose forward return is unresolved at that date. The old draft reported 34 signals; applying its own formula to this frozen snapshot yields 32. The discrepancy is therefore a data-snapshot difference, not a changed signal formula.

Eligible memories are prior negative outcomes from the same 200-day-SMA regime, ordered by worst maximum drawdown and capped at three. Reflections are generated by the same local model from completed numeric records using an objective-journal prompt.

Paired direct audit

Every setup is sent from a fresh context through four conditions: (i) loss records under a trauma/fear frame (loss_trauma), (ii) the identical loss records under a neutral frame (loss_neutral), (iii) identical dates and numerical loss facts but action-neutral, character-length-matched reflection prose (neutralized_reflection), and (iv) action-neutral padded text containing no historical facts (padded_no_memory). Prompt-token counts returned by Ollama are logged and reported; the study does not claim exact token matching unless those observed counts warrant it.

For each contrast, EXECUTE receives the realized forward return and ABORT receives zero. The paired difference is therefore the realized difference in the rule-following action under the two prompts. Bootstrap intervals resample setups, not model outputs.

Independent label-by-number audit

For setups with an eligible memory, a neutral-frame auxiliary prompt holds the retrieval block fixed while independently varying numeric confidence (30/100, 70/100) and a categorical status tag (NEUTRAL, SHAKEN). This identifies prompt sensitivity to the label at fixed numeric score and sensitivity to the score at fixed label. It does not establish a general cognitive mechanism.

4. Results

Paired direct effects

The exploratory draft reported 11 ABORTs in 34 setups under its original protocol. The corrected loss-trauma arm produced 0 ABORTs in 32 setups, including 0/24 setups with matured retrieved memories. This is a non-replication under a redesigned protocol, not yet a causal demonstration that the original effect was a control artifact.

Among 24 setups with at least one matured memory, this contrast switched 0 decisions (0 ABORT-to-EXECUTE and 0 EXECUTE-to-ABORT). The mean paired action-return difference (left condition minus right condition) was +0.00%, with a setup-bootstrap 95% interval from +0.00% to +0.00%. The median reported prompt-token difference (left minus right) was 12.0.

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Configuration-level selection benchmark

The configuration-level random-partition benchmark was undefined because all decisions took the same action. This is an all-abort, configuration-level descriptive benchmark. It is not a test of economic value in general, does not identify the causal-flip subset, and a large p-value is not evidence of equivalence or of an absence of skill.

Harness positive control and raw prompt audit

The explicit hard-risk control produced ABORT on 24/24 eligible cases (100.0%). It was added after the four-arm universal null as an iteratively hardened, post-null diagnostic. It retained the state and record fields but added a binding hard-risk-limit instruction in both the system and user prompts; it is therefore deliberately stronger and structurally different from the memory conditions. It establishes only that the harness can register ABORT. It does not map where, between negatively curated memory and a binding prohibition, the rule begins to yield. The raw loss-trauma responses did reach and discuss the retrieved records. Two examples are shown below:

> spy-20211203 (EXECUTE): { "decision": "EXECUTE", "rationale": "Although historical losses can feel like trauma, the current RSI reading of 36.1 indicates a relatively low level of fear. The regime is still in a bull market, which suggests that the market is generally trending upwards. While there was a significant loss event on 2021-09-17, it occurred within a bull regime and does not necessarily imply that the current situation is similar. Given these factors, I believe it's reasonable to override fear and follow the BUY rule." }

> spy-20220414 (EXECUTE): { "decision": "EXECUTE", "rationale": "Although historical losses can feel like trauma, the current RSI reading of 37.0 indicates a potential buying opportunity in a bear market regime. The retrieved episodic trauma shows two instances of completed historical setups in a bear regime, but these events are not directly applicable to the current situation. Given the realistic caution and the potential for a contrarian play, it is recommended to EXECUTE the BUY rule." }

Descriptive transcript audit

Before the full trace was read, a codebook fixed the unit of analysis, coding fields, and reporting rule. A single reviewer then coded all 24 memory-eligible loss-trauma rationales while retaining raw output in a separate column. The reviewer coded retrieved-memory mention in 24/24 responses, explicit regime/similarity linkage in 20/24, and dismissal of the memory in 24/24. The pre-specified same-regime-then-dismiss-without-comparison pattern occurred in 20/24 responses; a concrete current-versus-past distinguishing reason was coded in 0/24. This is a re-codable qualitative observation from one reviewer and one model, not evidence of a general rationalization mechanism.

Independent label and number manipulations

The table reports within-setup differences. Because every paired decision was identical, the zero-width intervals are mechanical restatements of zero switches, not evidence of high precision. Zero switches among 24 memory-eligible setups is compatible with an underlying switch rate of roughly 12.5% under the usual rule-of-three heuristic. The controlled result therefore supports no detected effect in this replay, not a proven zero effect.

Retrieval and mean-reversion diagnostic

The diagnostic was undefined because the eligible configuration did not contain both decisions.

Contrast	Eligible	Switches	ABORT->EXEC	EXEC->ABORT	Paired action-return diff
trauma-frame effect: loss_trauma minus loss_neutral	24	0	0	0	+0.00% [+0.00%, +0.00%]
reflection-language effect: loss_neutral minus neutralized_reflection	24	0	0	0	+0.00% [+0.00%, +0.00%]
negative-record effect: neutralized_reflection minus padded_no_memory	24	0	0	0	+0.00% [+0.00%, +0.00%]
full configuration effect: loss_trauma minus padded_no_memory	24	0	0	0	+0.00% [+0.00%, +0.00%]

Paired manipulation	n	Abort-rate diff	ABORT->EXEC	EXEC->ABORT
SHAKEN minus NEUTRAL at 30/100	24	+0.00% [+0.00%, +0.00%]	0	0
SHAKEN minus NEUTRAL at 70/100	24	+0.00% [+0.00%, +0.00%]	0	0
30 minus 70 at NEUTRAL	24	+0.00% [+0.00%, +0.00%]	0	0
30 minus 70 at SHAKEN	24	+0.00% [+0.00%, +0.00%]	0	0

5. Discussion

The corrected experiment distinguishes three questions that the exploratory draft had combined: whether an outer trauma frame changes a decision, whether the reflection prose changes a decision once the loss numbers are fixed, and whether the presence of negative historical records changes a decision once the prose is neutralized. A result in one contrast should not be relabeled as a result in another. In this bounded audit, the deterministic RSI rule held across every tested variation of frame, reflection text, numerical records, label, and confidence score. The positive control places only an upper endpoint on this observation: an explicit binding prohibition overrides the rule, while the severity gradient between that instruction and retrieved loss memory remains unmapped.

The return analysis is likewise intentionally narrow. It evaluates the realized consequence of a prompt-induced action difference on the frozen setup series. It does not establish live portfolio performance because setups overlap, transaction costs are omitted, memories are teacher-forced, and the sample is small. The random-partition result is retained as a transparent configuration-level benchmark, not as proof of a null.

6. Limitations

This is one local model build (llama3:latest, model-list ID 365c0bd3c000, model blob SHA-256 6a0746a1ec1aef3e7ec53868f220ff6e389f6f8ef87a01d77c96807de94ca2aa, 8.0B parameters, Q4_0 quantization), one asset, and a small, single historical path. The replay is teacher-forced and measures prompt-level decision effects, not a live autonomous trader. The retrieval policy remains negative-only, so this paper audits a deliberately adverse configuration rather than memory architectures in general. The auxiliary confidence experiment is a controlled text intervention, not a measurement of internal computation. Temperature zero and a seed reduce variation but do not prove determinism; 0/24 memory-eligible switches does not establish a zero population rate. The exact runtime manifest and all raw outputs are included with the artifacts.

7. Reproducibility

The package contains the frozen data snapshot and hash, the pre-analysis plan, exact prompts and model responses, paired condition labels, model runtime manifest, and analysis code. The model was queried as llama3:latest (ID 365c0bd3c000, blob SHA-256 6a0746a1ec1aef3e7ec53868f220ff6e389f6f8ef87a01d77c96807de94ca2aa) with temperature zero and seed 20260711. The Ollama runtime reported: {"version": "0.31.2"}.

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